INCH-POUND

MS51813F
w/AMENDMENT 1
10 October 2018
SUPERSEDING
MS51813F
3 April 2018

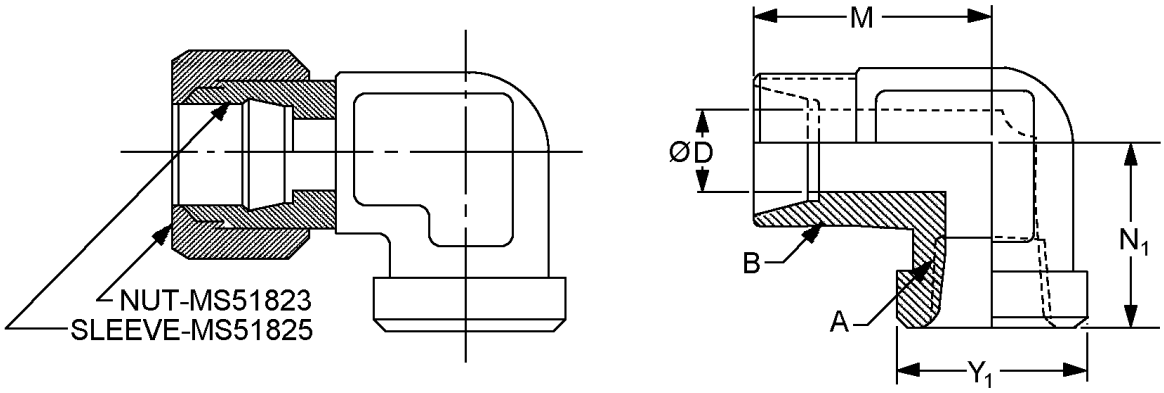
DETAIL SPECIFICATION SHEET

ELBOW, PIPE TO TUBE 90 DEGREE, FEMALE PIPE END, FLARELESS TYPE, HYDRAULIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Reactivated after 3 April 2018 and may be used for new and existing designs and acquisitions.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-18866.



See notes at end of figure.

FIGURE 1. Elbow pipe to tube.

AMSC N/A

FSC 4730

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Standard sizes Dash number		Tube OD Nom inches (mm)	A Thread NPTF	B Straight thread Class 2A	D dia. Drill inches (mm)
Assembly	Body (see note 5)				
1	21	.1250 (3.175)	.125 -27	.3125 - 24UNF	.093 (2.36)
2	22	.1875 (4.763)	.125 -27	.3750 - 24UNF	.125 (3.18)
3	23	.2500 (6.350)	.125 -27	.4375 - 20UNF	.203 (5.16)
4	24	.3125 (7.936)	.125 -27	.5000 - 20UNF	.234 (5.94)
5	25	.3750 (9.525)	.250 - 18	.5625 - 18UNF	.281 (7.14)
6	26	.5000 (12.700)	.375 - 18	.7500 - 16UNF	.422 (10.72)
7	27	.6250 (15.875)	.500 - 14	.8750 - 14UNF	.500 (12.70)
8	28	.7500 (19.050)	.750 - 14	1.0625 - 12UN	.656 (16.66)
9	29	.8750 (22.225)	.750 - 14	1.1875 - 12UN	.718 (18.24)
10	30	1.0000 (25.400)	1.000 - 11 1/2	1.3125 - 12UN	.875 (22.23)
11	31	1.2500 (31.750)	1.250 - 11 1/2	1.6250 - 12UN	1.093 (27.76)
12	32	1.5000 (38.100)	1.500 - 11 1/2	1.8750 - 12UN	1.344 (34.14)
13	33	2.0000 (50.800)	2.000 - 11 1/2	2.5000 - 12UN	1.813 (46.05)

Standard sizes Dash number		M_1 inches (mm) ±.03 (0.76)	N_1 inches (mm) ±.03 (0.76)	Y_1 inches (mm) ±.02(0.51)
Assembly	Body (see note 5)			
1	21	.83 (21.08)	.66 (16.76)	.562 (14.27)
2	22	.83 (21.08)	.66 (16.76)	.562 (14.27)
3	23	.89 (22.61)	.66 (16.76)	.562 (14.27)
4	24	.95 (24.13)	.66 (16.76)	.562 (14.27)
5	25	1.05 (26.67)	.88 (22.35)	.750 (19.05)
6	26	1.23 (31.24)	1.02 (25.91)	.875 (22.23)
7	27	1.42 (36.07)	1.23 (31.24)	1.062 (26.97)
8	28	1.58 (40.13)	1.36 (34.54)	1.312 (33.32)
9	29	1.62 (41.15)	1.42 (36.07)	1.312 (33.32)
10	30	1.73 (43.94)	1.62 (41.15)	1.625 (41.28)
11	31	2.08 (52.83)	1.70 (43.18)	1.875 (47.63)
12	32	2.58 (65.53)	2.08 (52.83)	2.562 (65.07)
13	33	2.64 (67.06)	2.39 (60.71)	2.812 (71.42)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Remove all burrs and slivers.
4. Dimensions and tolerances not shown shall be in accordance with SAE-J514.
5. Assemblies are only furnished to this specification (body, nut, and sleeve). Bodies are not to be stocked, stored, or issued.
6. The drawing is for identification purposes only and is not intended to restrict designs and shapes not dimensioned.

FIGURE 1. Elbow pipe to tube - Continued.

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REQUIREMENTS:

Elbow, pipe to tube shall be as specified on figure 1 and in table I.
Materials shall be in accordance with MIL-DTL-18866.

Finish. Finishes shall be as specified in table I. All plating's shall be capable of meeting a minimum of 96 hours salt spray test in accordance with ASTM B117. The fittings shall show no evidence of red corrosion after 96 hours of salt spray. Fluid passages, other openings and internal threads shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.

TABLE I. Material and finish identification codes.

PIN code material/plating finish	Material	Plating finish
Blank	Steel	Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2. 1/
H	Steel	Zinc-Aluminum in accordance with ASTM F1136/F1136M, grade 3, NC. 2/
J	Steel	Zinc-nickel in accordance with SAE-AMS2417, type 2, grade B. 2/ 3/
M	Nickel-copper alloy UNS N04400	No additional finish.
N	High-chromium nickel alloy UNS N06690	No additional finish.
P	Steel	Zinc phosphate finish in accordance MIL-DTL-16232 type Z, class 1. 4/
R	Steel	Zinc plating in accordance with ASTM B633; type III or V, Fe/Zn 8. 4/ 5/
SS	Corrosion resistant steel	No additional finish. Passivation in accordance with SAE-AMS2700, method 1, type 6 or 7, free machining alloy UNS S30300 method 1, type 2 or 4.
T	Titanium	Anodize in accordance with SAE-AMS2488 type 2. 6/
TF	Titanium	Fluoride phosphate in accordance with SAE-AMS2486. 6/
Z	Steel	Zinc plating in accordance with ASTM B695, type II, class 8. 4/ 5/
ZC	Steel	Zinc may be any zinc plating from PIN codes J, R, and Z with a colored chromate coating. 4/ 5/

1/ Embrittlement test need not be run. Cadmium shall not be used in oxygen or potable water systems. Cadmium is prohibited from use in NAVSEA owned shipboard systems.
2/ Hexavalent chromium free. Finish shall be RoHS compliant.
3/ The zinc-nickel alloy plate shall contain 12% to 16% nickel. The coating thickness shall be 315µ inches (8µm) minimum.
4/ Not for use in aircraft. Requires approval from the Program Office for all applications. Zinc-plated materials should be avoided in hydraulic systems with water-glycol fluids. Reference ASTM MNL37 - Fuels and Lubricants Handbook, "water glycol fluids are highly alkaline due to the presence of amine based corrosion inhibitors. As a result, these fluids can attack zinc, cadmium, magnesium, and non-anodized aluminum, forming sticky or gummy residues. Consequently, these metals should be avoided when selecting system components."

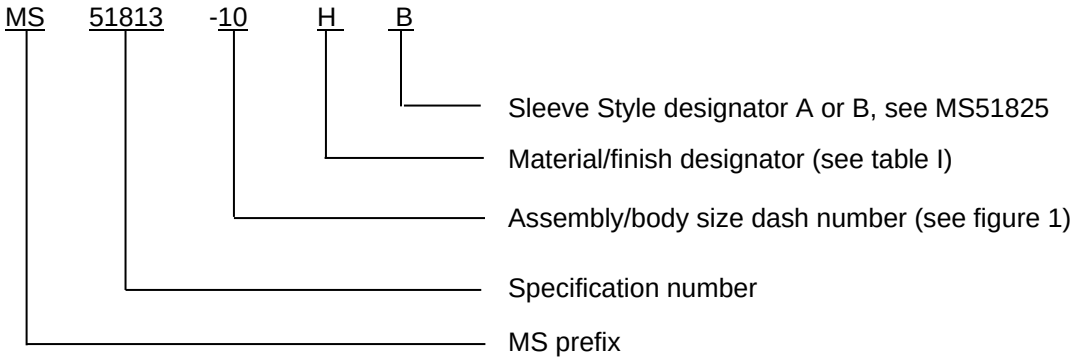
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5/ These finishes have a chromate conversion coating which includes hexavalent chromium and are not recommended.

6/ A pretreatment, a modification of the fluoride treatment, or a post treatment shall be applied so the final color of the fittings shall be similar to SAE-AMS-STD-595 colors 36076 through 36293. Titanium shall not be used in oxygen or potable water systems.

Maximum operating pressure. Maximum operating pressure shall be in accordance with SAE-J514.

PIN: The PIN consists of the letters “MS”, the specification number, a dash, number for an elbow pipe to tube 90° assembly/body size, and a letter for material finish designator.



PIN example: MS51813-10H indicates a 90° elbow, pipe to tube assembly, 1.000 inch (2.540 mm) tube OD, with 1.3125 - 12UN threads, zinc-aluminum.

Cadmium is not recommended. To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when other materials and finishes specified in this document cannot meet performance requirements.

Cadmium is prohibited from use in NAVSEA owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2.

Class I and II ozone depleting substances (ODS) shall not be used in MS51813 or any referenced procedures.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Table II provides a detail reference data listing for MS39200 to MS51813. This table contains coordination of equivalent tube sizes, in both compression and hydraulic type elbows. Assembly replacement is authorized.

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TABLE II. Assembly equivalency (for information only).

PIN		PIN		PIN	
Compression MS39200	Hydraulic MS51813	Compression MS39200	Hydraulic MS51813	Compression MS39200	Hydraulic MS51813
-1	-1	-5	-5	-9	-9
-2	-2	-6	-6	-10	-10
-3	-3	-7	-7	-11	-11
-4	-4	-8	-8	-12	-12

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-18866, this document references the following:

MIL-DTL-16232
MS51823
MS51825
ASTM B117
ASTM B633
ASTM B695
ASTM F1136/F1136M
ASTM MNL37
SAE-AMS-C-81562
SAE-AMS-QQ-P-416
SAE-AMS-STD-595/36076
SAE-AMS-STD-595/36081
SAE-AMS-STD-595/36099
SAE-AMS-STD-595/36118
SAE-AMS-STD-595/36134
SAE-AMS-STD-595/36152
SAE-AMS-STD-595/36170
SAE-AMS-STD-595/36173
SAE-AMS-STD-595/36176
SAE-AMS-STD-595/36231
SAE-AMS-STD-595/36251
SAE-AMS-STD-595/36270
SAE-AMS-STD-595/36280
SAE-AMS-STD-595/36293
SAE-AMS2417
SAE-AMS2486
SAE-AMS2488
SAE-AMS2700
SAE-J514
T9070-AL-DPC-020/077-2

CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - SH
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2018-107)

Review activities:
Army - AR, MI
Navy - AS, CG, SA
Air Force - 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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